

Desktop LED Emergency Light

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You can make an ultra-simple desktop emergency light using 'discharged' batteries. You know that battery cells still have considerable energy left in them even when these become 'weak' for most



Fig. 1: Author's prototype of desktop LED emergency light



Fig. 2: 1W white LED

battery-powered devices. Here is a way to extract their leftover power for use in such applications as

emergency light.

The LED light described here can be powered off any 1.5V or even a drained 1.2V battery cell. Fig. 1 shows author's prototype and Fig. 2 shows the 1W white LED used.

Circuit and working

Fig. 3 shows the circuit diagram of desktop LED emergency light that comprises a combination of a joule thief circuit, a 1W white LED chip and a blackout detector section.

The front end of the circuit is a blackout detector for sensing the presence and absence of the grid power. Here, optocoupler PC817 (IC1) is connected to 230V AC mains supply through capacitor C1, diode D1, and resistors R1, R2 and R3.

PARTS LIST

Semiconductors:

- IC1 - PC817 optocoupler
- T1, T2 - BC548 npn transistor
- D1 - 1N4007 rectifier diode
- ZD1 - 5.1V, 0.5W zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 470-kilo-ohm
- R2 - 100-ohm
- R3, R4 - 1-kilo-ohm

Capacitors:

- C1 - 100nF, 400V polyester
- C2 - 220 μ F, 16V electrolytic
- C3 - 10nF ceramic disk

Miscellaneous:

- X1 - 10mm ferrite core toroidal transformer
- S1 - On/off switch
- CON1, CON2 - 2-pin terminal connector
- CON3 - 2-pin connector
- 1.5V (used) battery
- 1W white LED

Voltage derived from AC mains is filtered by capacitor C2 and regulated by zener diode ZD1. This circuit section is used to control the switching of the joule thief circuit built around transistor T2, capacitor C3, resistor R4, and inductors L1 and L2. Resistor R4 limits the base current of transistor T2 and capacitor C3 improves the circuit efficacy.

When switch S1 is closed and the battery is connected at CON2 (and AC mains supply is not available), current flows into the base of transistor T2, the voltage across its base and emitter rises and the transistor is switched on. This means that a larger current can now flow through primary winding L1 of transformer X1 and then through the transistor. This collector current generates a magnetic field in the toroidal transformer in a direction opposite to the field created by the base current through its secondary winding L2.

As soon as the primary current becomes greater than the secondary current, the voltage on the secondary winding reverses which, in turn, switches off transistor T2. The magnetic field collapses and white LED is switched on. This allows the 1W white LED to be lit from a single 1.5V/1.2V battery, which otherwise required a 3.2V DC source or more.

Test Points

Test point	Details
TP0	0V
TP1	1.5V when S1 is on
Across TP2 and TP3	Around 5.1V when AC mains is available

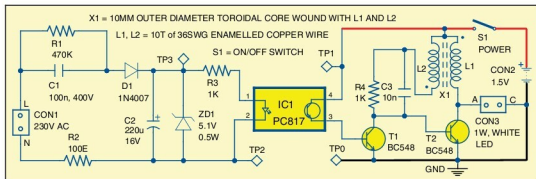


Fig. 3: Circuit diagram of the emergency light

Construction and testing

An actual-size, single-side PCB for the emergency light is shown in Fig. 4 and its component layout in Fig. 5. After assembling the circuit on PCB, enclose it in a suitable plastic box. When 230V AC

